Ensemble Methods

TD – PAC-Bayesian Learning

M2 Computer Science – MALIA

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Abstract. This tutorial accompanies Sections 2–7 of the lecture notes *PAC-Bayesian Learning: a Theoretical Framework for Ensemble Methods*. Exercises 1–3 are warm-ups (comprehension, classical short proofs, a first certificate by hand); Exercises 4–12 follow the notes; Exercises 13–17 mix applications and short proofs. Each title states the type of the exercise; those marked ($\star$) are at Master level. Notation: $S \sim \mathcal{D}^m$, voters $h : \mathcal{X} \to \{-1, +1\}$, posterior Q, prior P, Gibbs classifier G_Q, majority vote B_Q, $W_Q(x, y) = \mathbb{E}_{h\sim Q}\mathbb{1}[h(x) \neq y]$, joint error $e_Q = \mathbb{E}W_Q^2$, disagreement d_Q. Unless stated otherwise, $\delta = 0.05$, and $\overline{\mathrm{kl}}^{-1}(q, \varepsilon) = \sup\{p \in [q, 1] : \mathrm{kl}(q\|p) \leq \varepsilon\}$.

Part I – Warm-up

Exercise 1 – True or false? *(comprehension)*

For each statement, say whether it is true or false and justify your answer in one or two sentences (a short argument or a counterexample). Q and P are distributions on a set $\mathcal{H}$ of voters with values in $\{-1, +1\}$, and ties of the majority vote are counted as errors.

(a) For every posterior Q, the majority vote is at least as good as the Gibbs classifier: $R_{\mathcal{D}}(B_Q) \leq R_{\mathcal{D}}(G_Q)$.

(b) The empirical Gibbs risk $\widehat{R}_S(G_Q)$ always lies between the smallest and the largest empirical risk $\widehat{R}_S(h)$ of the voters h in the support of Q.

(c) The prior P may depend on domain knowledge or on data *other* than the sample S on which the bound is computed, but not on S itself.

(d) For the bound to be valid, the posterior Q must not depend on S either.

(e) Since a PAC-Bayesian bound holds for all posteriors simultaneously, we may compute it for a thousand posteriors and report the smallest value, without any correction.

(f) "With probability at least 0.95" means that at least 95% of the voters $h \sim P$ satisfy $R_{\mathcal{D}}(h) \leq$ the bound.

(g) For the same empirical risk $\hat{q}$ and the same complexity term ε, the kl-based bound $\overline{\mathrm{kl}}^{-1}(\hat{q}, \varepsilon)$ is never larger than the Hoeffding-type bound $\hat{q} + \sqrt{\varepsilon/2}$, and the gain is largest when $\hat{q}$ is close to 0.

(h) The disagreement d_Q can be estimated on unlabeled data, whereas the joint error e_Q needs labels.

(i) Taking $Q = P$ always gives the best bound, since it cancels the KL term.

(j) The tandem bound $4e_Q$ is always tighter than the first-order bound $2R_{\mathcal{D}}(G_Q)$.

Exercise 2 – From Markov to Hoeffding *(short proofs)*

In this exercise, Z is a real random variable with mean μ and variance v, and $a > 0$.

1. *(Markov).* If $Z \geq 0$, show that $\mathbb{P}(Z \geq a) \leq \frac{\mathbb{E}Z}{a}$ (hint: compare $a\mathbb{1}[Z \geq a]$ and Z). Deduce that for every $\delta \in (0, 1]$, $Z \leq \frac{\mathbb{E}Z}{\delta}$ with probability at least $1 - \delta$. Give a distribution for which Markov's inequality is an equality.

2. *(Chebyshev).* Show that $\mathbb{P}(|Z - \mu| \geq a) \leq \frac{v}{a^2}$.

3. *(Cantelli).* For $t \geq 0$, show that $\mathbb{P}(Z - \mu \geq a) \leq \frac{v+t^2}{(a+t)^2}$, then optimize over t to obtain $\mathbb{P}(Z - \mu \geq a) \leq \frac{v}{v+a^2}$. Compare the two inequalities numerically for $v = 0.01$ and $a = 0.2$, and explain why the one-sided bound is always better than Chebyshev's.

4. *(Union bound).* Show that $\mathbb{P}(\bigcup_{k=1}^{K} A_k) \leq \sum_{k=1}^{K} \mathbb{P}(A_k)$ for any events $A_1, \dots, A_K$. Deduce that if K statements hold each with probability at least $1 - \delta/K$, then they hold *simultaneously* with probability at least $1 - \delta$.

5. *(Hoeffding's inequality).* Admit Hoeffding's lemma: if $\mathbb{E}X = 0$ and $\alpha \leq X \leq \beta$, then $\mathbb{E}e^{sX} \leq e^{s^2(\beta-\alpha)^2/8}$ for all $s \in \mathbb{R}$. Let $Z_1, \dots, Z_m$ be i.i.d. in $[0, 1]$ with mean p and $\hat{p} = \frac{1}{m}\sum_i Z_i$.

   (a) Show that $\mathbb{E}e^{\lambda(p-\hat{p})} \leq e^{\lambda^2/(8m)}$ for every $\lambda > 0$.

   (b) *(Chernoff's method).* Apply Markov's inequality to $e^{\lambda(p-\hat{p})}$ and optimize over λ to show that $\mathbb{P}(p - \hat{p} \geq \varepsilon) \leq e^{-2m\varepsilon^2}$.

   (c) Deduce that, with probability at least $1 - \delta$, $p \leq \hat{p} + \sqrt{\ln(1/\delta)/(2m)}$, and give a two-sided version. Using question 4, give a bound valid simultaneously for K fixed classifiers.

6. ($\star$) *(Hoeffding's lemma).* Let $\alpha \leq X \leq \beta$ with $\mathbb{E}X = 0$ (so $\alpha < 0 < \beta$ unless $X = 0$). Using the convexity of $x \mapsto e^{sx}$, show that $\mathbb{E}e^{sX} \leq e^{L(u)}$ with $u = s(\beta - \alpha)$, $\theta = \frac{-\alpha}{\beta-\alpha}$ and $L(u) = -\theta u + \ln(1 - \theta + \theta e^u)$. Show that $L(0) = L'(0) = 0$ and $L'' \leq \frac{1}{4}$, and conclude.

Exercise 3 – Certifying one classifier *(application)*

A classifier f (for instance a random forest) has been learned on a sample S_1. It is then evaluated on a test sample T of $n = 200$ examples drawn i.i.d. from $\mathcal{D}$, independently of S_1, and makes 10 errors.

1. Why is $n\widehat{R}_T(f)$ a binomial variable $\mathrm{Bin}(n, R_{\mathcal{D}}(f))$ conditionally on S_1? Which assumption would break if T had been used to tune the hyperparameters of f?

2. Compute the Hoeffding test-set bound $\widehat{R}_T(f) + \sqrt{\ln(1/\delta)/(2n)}$.

3. The kl test-set bound is $\overline{\mathrm{kl}}^{-1}(\widehat{R}_T(f), \frac{\ln(1/\delta)}{n})$. Compute $\varepsilon = \ln(1/\delta)/n$, check that $\mathrm{kl}(0.05\|0.15) > \varepsilon$, and perform four steps of bisection on $[0.05, 0.15]$ (compute $\mathrm{kl}(0.05\|c)$ at each midpoint c). Compare the result with the Pinsker relaxation $\hat{q} + \sqrt{\varepsilon/2}$ and with the refined relaxation $\hat{q} + \sqrt{2\hat{q}\varepsilon} + 2\varepsilon$.

4. In fact, the engineer trained $K = 20$ classifiers (different hyperparameters), evaluated all of them on T and kept the one with the fewest test errors, which is f. Why are the certificates of questions 2 and 3 no longer valid? Correct them with a union bound, for $K = 20$ and $K = 1000$ (for the kl bound you may use `kl_inv_upper` of `pbtools`).

5. What test size n guarantees a Hoeffding width $\sqrt{\ln(K/\delta)/(2n)} \leq 0.02$, for $K \in \{1, 20, 1000\}$? Comment.

6. Show that the corrected bound of question 4 is the Occam bound (Section 2.4 of the notes) for a well-chosen prior, and give its KL term.

Part II – Core exercises

Exercise 4 – The Kullback–Leibler divergence *(short proofs)*

Let $\mathcal{H} = \{h_1, \dots, h_n\}$ be finite and P, Q two distributions on $\mathcal{H}$ with $P_i > 0$.

1. Using Jensen's inequality, show that $\mathrm{KL}(Q\|P) \geq 0$, with equality iff $Q = P$.
2. Assume P uniform. Show that $\mathrm{KL}(Q\|P) = \ln n - H(Q)$ where $H(Q) = -\sum_i Q_i \ln Q_i$. What are the minimal and maximal values of $\mathrm{KL}(Q\|P)$? Compute it when Q is uniform over $k \leq n$ voters.
3. Show that $\mathrm{KL}(Q \otimes Q \,\|\, P \otimes P) = 2\,\mathrm{KL}(Q\|P)$.
4. Show that $\mathrm{KL}\left(\mathcal{N}(\mu_1, \sigma^2 I_d) \,\|\, \mathcal{N}(\mu_0, \sigma^2 I_d)\right) = \frac{\|\mu_1 - \mu_0\|^2}{2\sigma^2}$.
5. ($\star$) Binary kl. Fix $q \in [0,1]$. Show that $p \mapsto \mathrm{kl}(q\|p)$ is convex on $(0,1)$, minimal at $p = q$, and prove Pinsker's inequality $\mathrm{kl}(q\|p) \geq 2(q-p)^2$ by studying $g(p) = \mathrm{kl}(q\|p) - 2(q-p)^2$.

Exercise 5 – Change of measure and Gibbs posterior *(short proofs)*

Let $\phi : \mathcal{H} \to \mathbb{R}$ and P a distribution on a finite set $\mathcal{H}$. Define $Q_\phi(h) = \frac{P(h)e^{\phi(h)}}{\mathbb{E}_P e^\phi}$.

1. Show that for any Q, $\mathrm{KL}(Q\|Q_\phi) = \mathrm{KL}(Q\|P) - \mathbb{E}_Q\phi + \ln \mathbb{E}_P e^\phi$.
2. Deduce the change of measure inequality $\mathbb{E}_Q\phi \leq \mathrm{KL}(Q\|P) + \ln \mathbb{E}_P e^\phi$ and the variational formula $\ln \mathbb{E}_P e^\phi = \max_Q \{\mathbb{E}_Q\phi - \mathrm{KL}(Q\|P)\}$.
3. Let $\lambda > 0$. Show that the minimizer of $Q \mapsto \widehat{R}_S(G_Q) + \frac{1}{\lambda m}\,\mathrm{KL}(Q\|P)$ is the Gibbs posterior $Q_\lambda(h) \propto P(h)e^{-\lambda m \widehat{R}_S(h)}$, and give the minimal value.
4. Describe Q_λ when $\lambda \to 0$ and $\lambda \to +\infty$. Interpret λ.
5. Recover Q_λ directly with a Lagrange multiplier for the constraint $\sum_h Q(h) = 1$.

Exercise 6 – From the general theorem to classical bounds *(proofs and application)*

Recall the general theorem: for convex Δ, w.p. $\geq 1-\delta$, for all Q, $\Delta(\widehat{R}_S(G_Q), R_{\mathcal{D}}(G_Q)) \leq \frac{1}{m}[\mathrm{KL}(Q\|P) + \ln \frac{1}{\delta}\mathbb{E}_{S'}\mathbb{E}_{h\sim P} e^{m\Delta(\widehat{R}_{S'}(h), R_{\mathcal{D}}(h))}]$.

1. Explain precisely where each of the following is used in the proof: convexity of Δ, the change of measure, Markov's inequality, the independence of P from S.
2. *(Hoeffding-type bound).* Take $\Delta(q,p) = \lambda(p-q)$ for a fixed $\lambda > 0$. Using Hoeffding's lemma, show that w.p. $\geq 1-\delta$, for all Q: $R_{\mathcal{D}}(G_Q) \leq \widehat{R}_S(G_Q) + \frac{\mathrm{KL}(Q\|P) + \ln\frac{1}{\delta}}{\lambda m} + \frac{\lambda}{8}$.
3. Minimize the right-hand side in λ. Why is it not allowed to plug this value into the bound? Propose a correct way to optimize λ and give the resulting price.
4. *(Catoni's bound).* Let $C > 0$ and $\mathcal{F}_C(p) = -\ln(1 - p(1-e^{-C}))$. Show that for a fixed voter with risk p and the 0-1 loss, $\mathbb{E}_{S'} e^{-Cm\widehat{R}_{S'}(h)} = (1 - p(1-e^{-C}))^m$. Deduce Catoni's bound.
5. Numerical application: $m = 1000$, $\delta = 0.05$, $\widehat{R}_S(G_Q) = 0.1$, $\mathrm{KL}(Q\|P) = 5$. Compute McAllester's bound $\widehat{R}_S + \sqrt{(\mathrm{KL} + \ln\frac{2\sqrt{m}}{\delta})/(2m)}$ and the Hoeffding-type bound for $\lambda \in \{0.1, 0.2\}$.

Exercise 7 – Occam's razor as a PAC-Bayesian bound *(comprehension)*

1. Let $\mathcal{H}$ be countable and $Q = \delta_h$ a Dirac on $h \in \mathcal{H}$. Compute $\mathrm{KL}(\delta_h \| P)$. What does Seeger's bound give for a single voter? Compare with the Occam bound of the lecture notes.
2. Why can we not use Dirac posteriors when $\mathcal{H}$ is uncountable (e.g. linear classifiers with a Gaussian prior)?
3. Let $\mathcal{H} = \{h_1, h_2, \dots\}$ be countable and $P(h_k) = \frac{6}{\pi^2 k^2}$. Check that P is a probability and give the complexity term of h_k. Interpret this prior in terms of *description length*.

Exercise 8 – Bounds on the risk of the majority vote *(proofs and application)*

We assume voters in $\{-1, +1\}$, 0-1 loss, and count ties as errors.

1. Show that $R_{\mathcal{D}}(B_Q) = \mathbb{P}(W_Q \geq \frac{1}{2})$ and $R_{\mathcal{D}}(G_Q) = \mathbb{E}W_Q$. Deduce $R_{\mathcal{D}}(B_Q) \leq 2R_{\mathcal{D}}(G_Q)$. Give an example of distribution of W_Q for which this inequality is an equality.
2. Show that $e_Q = \mathbb{E}[W_Q^2]$, $d_Q = \mathbb{E}[2W_Q(1 - W_Q)]$, and $R_{\mathcal{D}}(G_Q) = e_Q + \frac{1}{2}d_Q$. Show also that $d_Q \leq 2R_{\mathcal{D}}(G_Q)(1 - R_{\mathcal{D}}(G_Q))$.
3. Prove the tandem bound $R_{\mathcal{D}}(B_Q) \leq 4e_Q$. Show that it is better than the first-order bound iff $d_Q \geq R_{\mathcal{D}}(G_Q)$.
4. For $\mu < \frac{1}{2}$, show that $R_{\mathcal{D}}(B_Q) \leq \frac{\mathbb{E}(W_Q - \mu)^2}{(1/2 - \mu)^2}$. Find the optimal $\mu^\star$ and show that the optimal value is $\frac{\mathrm{Var}\, W_Q}{\mathrm{Var}\, W_Q + (1/2 - \mathbb{E}W_Q)^2}$ (Cantelli's inequality).
5. Deduce the C-bound $R_{\mathcal{D}}(B_Q) \leq 1 - \frac{(1 - 2R_{\mathcal{D}}(G_Q))^2}{1 - 2d_Q}$ (when $R_{\mathcal{D}}(G_Q) < \frac{1}{2}$).
6. *(Condorcet).* Q is uniform over n voters erring independently with probability p. Compute $R_{\mathcal{D}}(G_Q)$, e_Q, d_Q as functions of n and p. For $p = 0.3$, compute the exact risk of the vote and the three bounds for $n = 3$, then the C-bound for $n = 25$ and its limit when $n \to \infty$. Comment.

Exercise 9 – Linear classifiers with a Gaussian posterior *(proofs)*

Voters $h_{\mathbf{v}}(x) = \mathrm{sign}\langle \mathbf{v}, x \rangle$, prior $P = \mathcal{N}(\mathbf{0}, I_d)$, posterior $Q_{\mathbf{w}} = \mathcal{N}(\mathbf{w}, I_d)$.

1. Show that $B_Q(x) = \mathrm{sign}\langle \mathbf{w}, x \rangle$.
2. Show that $\mathbb{P}_{\mathbf{v} \sim Q_{\mathbf{w}}}(h_{\mathbf{v}}(x) \neq y) = \Phi\big(-\frac{y\langle \mathbf{w}, x \rangle}{\|x\|}\big)$ where Φ is the standard Gaussian CDF. Give $\widehat{R}_S(G_{Q_{\mathbf{w}}})$ and $\mathrm{KL}(Q_{\mathbf{w}} \| P)$.
3. Show that, for a fixed C, minimizing Catoni's bound over $\mathbf{w}$ amounts to minimizing $C \sum_i \Phi(-y_i \langle \mathbf{w}, x_i \rangle / \|x_i\|) + \frac{1}{2}\|\mathbf{w}\|^2$. Compare with the SVM.
4. For $\mathbf{w} = s\mathbf{u}$ with $\|\mathbf{u}\| = 1$, describe the evolution of $\widehat{R}_S(G_{Q_{s\mathbf{u}}})$ and of the KL when s grows. Relate to the notion of margin.

Exercise 10 – (⋆) The PAC-Bayes-λ bound and its minimization *(proofs)*

1. Prove the refined Pinsker inequality $\mathrm{kl}(q \| p) \geq \frac{(p - q)^2}{2p}$ for $q < p$.
2. Starting from Seeger's bound, show that w.p. $\geq 1 - \delta$, for all Q and all $\lambda \in (0, 2)$ simultaneously, $R_{\mathcal{D}}(G_Q) \leq \frac{\widehat{R}_S(G_Q)}{1 - \lambda/2} + \frac{\mathrm{KL}(Q \| P) + \ln \frac{2\sqrt{m}}{\delta}}{\lambda(1 - \lambda/2)m}$. Why is no union bound over λ needed here, contrary to Exercise 6?
3. For a fixed Q, compute the optimal λ. For a fixed λ, compute the optimal Q on a finite set of voters. Write the alternating minimization algorithm.

4. Explain why the posterior returned by this algorithm on a random forest may give a *worse* majority vote than the uniform one.

Exercise 11 – (⋆) Out-of-bag samples *(proofs)*

Consider bagging with bootstrap samples of size k drawn with replacement among the m training examples.

1. Compute the probability that a given example is out-of-bag for a given tree; give its limit when $m \to \infty$ for $k = m$ and for $k = m/2$. Same question for being out-of-bag for two given trees.
2. Explain why the OOB loss $\widehat{L}_t$ of tree t is an unbiased estimate of $R_{\mathcal{D}}(h_t)$ given S_t, and why a uniform prior over the *indices* of the trees is a legitimate PAC-Bayesian prior although the trees depend on S.
3. Show that if $\mathbb{E}e^{N\,\mathrm{kl}(\hat{p}_N\|p)} \le 2\sqrt{N}$ for all N, then for $n_0 \le N$, $\mathbb{E}e^{n_0\,\mathrm{kl}(\hat{p}_N\|p)} \le 2\sqrt{n_0}$. Why is this useful?

Exercise 12 – (⋆) Self-bounding learning *(comprehension and proofs)*

Let $\mathcal{B}(Q, S, \delta)$ be a bound such that $\mathbb{P}_S(\forall Q,\ R_{\mathcal{D}}(Q) \le \mathcal{B}(Q, S, \delta)) \ge 1 - \delta$.

1. Let Q_S be the posterior returned by any algorithm run on S. Show that $\mathbb{P}_S(R_{\mathcal{D}}(Q_S) \le \mathcal{B}(Q_S, S, \delta)) \ge 1 - \delta$. Why does this fail for a bound that holds only for a *fixed* predictor (e.g. Hoeffding's inequality)?
2. You build $K = 5$ random forests with tree depths $1, 2, 4, 8$ and unlimited, and compute for each of them the OOB tandem certificate. How must you modify the certificates to be allowed to keep the forest with the smallest one? What is the cost in the complexity term?
3. Among the following choices, which are free, which require a union bound, which are forbidden? (a) the temperature λ of a Gibbs posterior; (b) the parameter C of Catoni's bound; (c) the variance of a Gaussian prior chosen on a grid of 10 values; (d) the centre of a Gaussian prior learned on the sample used in the bound; (e) the number of gradient iterations when minimizing the bound.
4. Let $p = \overline{\mathrm{kl}}^{-1}(q, \varepsilon)$. Show that $\frac{\partial p}{\partial \varepsilon} = \frac{p(1-p)}{p-q}$ and $\frac{\partial p}{\partial q} = \frac{p(1-p)}{p-q} \ln \frac{p(1-q)}{q(1-p)}$. Deduce the gradient of the first-order certificate $2\,\overline{\mathrm{kl}}^{-1}(\rho^\top \widehat{L}, \frac{\mathrm{KL}(\rho\|\pi)+c}{n})$ with respect to ρ.
5. Why is the test-set bound usually tighter than a self-bounding certificate for a random forest? Give one advantage of the self-bounding approach.

Part III – Applications and short proofs

Exercise 13 – A majority vote by hand *(application)*

Five voters $h_1, \dots, h_5$ are evaluated on eight labelled points. Their predictions are:

point i	1	2	3	4	5	6	7	8
label y_i	+	−	+	+	−	+	−	−
$h_1(x_i)$	−	+	+	+	−	+	−	−
$h_2(x_i)$	−	−	−	+	−	+	−	−
$h_3(x_i)$	−	−	+	−	−	+	−	−
$h_4(x_i)$	+	−	+	+	+	−	−	−
$h_5(x_i)$	+	−	+	+	−	+	+	+

We treat the eight points as the whole distribution $\mathcal{D}$ (uniform on the eight points), so that all quantities below are exact ("oracle") values. Ties of the vote count as errors.

1. Give the risk of each voter. For the uniform posterior $Q = (\frac{1}{5}, \dots, \frac{1}{5})$, compute $W_Q(x_i, y_i)$ and the margin $M_Q = 1 - 2W_Q$ on each point, then $R_{\mathcal{D}}(G_Q)$ and $R_{\mathcal{D}}(B_Q)$.
2. Compute the joint error $e_Q = \mathbb{E}W_Q^2$ and the disagreement $d_Q = \mathbb{E}[2W_Q(1-W_Q)]$, and check that $R_{\mathcal{D}}(G_Q) = e_Q + \frac{1}{2}d_Q$. Compute the 5×5 matrix D of pairwise disagreements $D_{jk} = \mathbb{P}_x(h_j(x) \neq h_k(x))$ and check that $d_Q = \sum_{j,k} Q_j Q_k D_{jk}$. Which of these computations need the labels?
3. Compute the first-order bound $2R_{\mathcal{D}}(G_Q)$, the tandem bound $4e_Q$ and the C-bound $1 - \frac{(1-2R_{\mathcal{D}}(G_Q))^2}{1-2d_Q}$, and compare them with $R_{\mathcal{D}}(B_Q)$. Give the optimal parameter $\mu^\star$ of the Chebyshev–Cantelli family (Exercise 8, question 4) and check that it gives the C-bound.
4. Same questions for the posterior $Q' = (0.1, 0.1, 0.1, 0.35, 0.35)$. Why do the second-order bounds improve while the first-order bound does not move?
5. Same questions for the Dirac posterior $Q'' = \delta_{h_1}$. Which bound is the best now? Comment.

Exercise 14 – The Gibbs posterior on four voters *(application)*

Four voters have, on a sample of size $m = 200$, the empirical risks $\widehat{R}_S(h_1) = 0.10$, $\widehat{R}_S(h_2) = 0.12$, $\widehat{R}_S(h_3) = 0.15$, $\widehat{R}_S(h_4) = 0.30$ (i.e. 20, 24, 30 and 60 errors). The prior P is uniform and $\delta = 0.05$. Recall the Gibbs posterior $Q_\lambda(h) \propto P(h)e^{-\lambda m \widehat{R}_S(h)}$ (Exercise 5), Seeger's bound $R_{\mathcal{D}}(G_Q) \leq \overline{\mathrm{kl}}^{-1}(\widehat{R}_S(G_Q), \frac{\mathrm{KL}(Q\|P)+\ln(2\sqrt{m}/\delta)}{m})$ and McAllester's bound $R_{\mathcal{D}}(G_Q) \leq \widehat{R}_S(G_Q) + \sqrt{\frac{\mathrm{KL}(Q\|P)+\ln(2\sqrt{m}/\delta)}{2m}}$. You may use `kl_inv_upper` of `pbtools` for the inversions.

1. Show that $Q_\lambda(h_j) = \frac{e^{-\lambda(k_j-k_1)}}{\sum_l e^{-\lambda(k_l-k_1)}}$, where k_j is the number of errors of h_j. Why is this form numerically preferable? Compute Q_λ for $\lambda \in \{0.05, 0.25, 1\}$.
2. For $\lambda \in \{0, 0.05, 0.25, 1\}$ and for $\lambda \to +\infty$, compute $\widehat{R}_S(G_{Q_\lambda})$, $\mathrm{KL}(Q_\lambda\|P)$ (use $\mathrm{KL} = \ln 4 - H(Q_\lambda)$, Exercise 4), and both bounds.
3. For $\lambda = 0.25$, compute the objective $\widehat{R}_S(G_Q) + \frac{\mathrm{KL}(Q\|P)}{\lambda m}$ for Q_λ, for the uniform posterior and for δ_{h_1}, and check the value $-\frac{1}{\lambda m}\ln \mathbb{E}_{h\sim P} e^{-\lambda m \widehat{R}_S(h)}$ found in Exercise 5.
4. Which posterior gives the best certificate? Is it legitimate to choose λ after computing the bounds? Why does the bound favour concentrated posteriors here, and what would this imply for the majority vote?

Exercise 15 – Data-dependent priors and sample splitting *(short proof and application)*

We learn a linear classifier with a Gaussian posterior $Q = \mathcal{N}(\mathbf{w}, I_d)$ from a sample S of size $m = 1000$, $\delta = 0.05$. To obtain a better prior, we split $S = S_1 \cup S_2$ with $|S_1| = m_1$ and $|S_2| = m_2 = m - m_1$, learn a vector $\mathbf{w}_{S_1}$ on S_1 only ($\mathbf{w}_{S_1} = \mathbf{0}$ if $m_1 = 0$), and use the prior $P_{S_1} = \mathcal{N}(\mathbf{w}_{S_1}, I_d)$. The posterior centre $\mathbf{w}$ is learned on the whole sample S.

1. *(Short proof.)* Show that, with probability at least $1-\delta$ over (S_1, S_2), for all posteriors Q, $R_{\mathcal{D}}(G_Q) \leq \overline{\mathrm{kl}}^{-1}(\widehat{R}_{S_2}(G_Q), \frac{\mathrm{KL}(Q\|P_{S_1})+\ln(2\sqrt{m_2}/\delta)}{m_2})$. Why may Q depend on the whole of S, while the empirical risk must be computed on S_2 only?
2. The following values were observed for four splits (in all cases $\widehat{R}_{S_2}(G_Q) = 0.10$):

m_1	0	200	500	800
$\|\mathbf{w} - \mathbf{w}_{S_1}\|^2$	120	30	10	4

Compute the KL (Exercise 4) and the certificate for each split.

3. Discuss the trade-off governing the choice of m_1.
4. An analyst computes the four certificates and reports the smallest one. What should she report instead?
5. What does the bound become if the posterior is the prior itself, $Q = P_{S_1}$? Relate it to the test-set bound of Exercise 3.

Exercise 16 – Certifying the balanced error *(short proof and application)*

For a classifier f, let $R_+(f) = \mathbb{P}(f(x) \neq y \,|\, y = +1)$ and $R_-(f) = \mathbb{P}(f(x) \neq y \,|\, y = -1)$ be the class-conditional risks, $\mathrm{BER}(f) = \frac{1}{2}(R_+(f) + R_-(f))$ the balanced error, and $\pi_+ = \mathbb{P}(y = +1)$, so that $R_{\mathcal{D}}(f) = \pi_+ R_+(f) + (1 - \pi_+)R_-(f)$. A classifier f, learned on another sample, is evaluated on an independent test sample of $n = 1000$ examples with $n_+ = 100$ positives and $n_- = 900$ negatives; it makes 12 errors on the positives and 18 on the negatives.

1. Compute the kl test-set bound on $R_{\mathcal{D}}(f)$. Why is this certificate misleading for this problem?
2. *(Short proof.)* Conditionally on the labels $(y_1, \dots, y_n)$, show that $n_+ \widehat{R}_+(f) \sim \mathrm{Bin}(n_+, R_+(f))$, where $\widehat{R}_+(f)$ is the error rate on the positive test examples. Deduce that with probability at least $1 - \delta$, simultaneously, $R_+(f) \leq U_+ := \overline{\mathrm{kl}}^{-1}(\widehat{R}_+(f), \frac{\ln(2/\delta)}{n_+})$ and $R_-(f) \leq U_- := \overline{\mathrm{kl}}^{-1}(\widehat{R}_-(f), \frac{\ln(2/\delta)}{n_-})$; hence $\mathrm{BER}(f) \leq \frac{U_+ + U_-}{2}$ and $R_{\mathcal{D}}(f) \leq \max(U_+, U_-)$.
3. Compute U_+, U_- and the certificate on the balanced error. What does it add to question 1? What could you say about $R_{\mathcal{D}}(f)$ if $\pi_+ = 0.1$ were known?
4. ($\star$) *(Another route.)* Conditionally on the labels, write the empirical balanced error as a sum $\sum_i Z_i$ of independent variables with $Z_i \in [0, c_i]$, $c_i = \frac{1}{2n_{y_i}}$. Adapting the Chernoff method of Exercise 2, show that $\mathrm{BER}(f) \leq \widehat{\mathrm{BER}}(f) + \sqrt{\frac{\ln(1/\delta)}{8}(\frac{1}{n_+} + \frac{1}{n_-})}$ with probability $\geq 1 - \delta$. Compute it and compare with question 3.
5. How would you obtain a PAC-Bayesian certificate on the balanced error of the Gibbs classifier G_Q, valid for all posteriors Q? And of the majority vote?

Exercise 17 – Softmax parametrization and the gradient of a certificate *(short proofs and application)*

A posterior ρ on n_v voters is parametrized by logits $\theta \in \mathbb{R}^{n_v}$: $\rho = \mathrm{softmax}(\theta)$, i.e. $\rho_t = e^{\theta_t} / \sum_s e^{\theta_s}$. Let π be the prior, $\widehat{L} \in [0,1]^{n_v}$ the vector of empirical losses, $\mathcal{B}(\rho)$ a differentiable objective and $g = \nabla_\rho \mathcal{B}$.

1. Show that $\frac{\partial \rho_t}{\partial \theta_s} = \rho_t(\mathbb{1}[t = s] - \rho_s)$, i.e. the Jacobian is $\mathrm{diag}(\rho) - \rho\rho^\top$.
2. Deduce $\nabla_\theta \mathcal{B} = \rho \odot (g - (\rho^\top g)\mathbf{1})$. Show that $\mathbf{1}^\top \nabla_\theta \mathcal{B} = 0$ and explain this with the invariance $\mathrm{softmax}(\theta + c\mathbf{1}) = \mathrm{softmax}(\theta)$. Why can g be modified by adding a multiple of $\mathbf{1}$?
3. Show that $\nabla_\rho \mathrm{KL}(\rho\|\pi) = \ln \frac{\rho}{\pi} + \mathbf{1}$ and $\nabla_\theta \mathrm{KL}(\rho\|\pi) = \rho \odot (\ln \frac{\rho}{\pi} - \mathrm{KL}(\rho\|\pi)\mathbf{1})$. What is this gradient at $\rho = \pi$?
4. Let $F(\rho) = \rho^\top \widehat{L} + \frac{\mathrm{KL}(\rho\|\pi)}{\lambda n}$. Show that $\nabla_\theta F = 0$ iff ρ is the Gibbs posterior $\rho_t \propto \pi_t e^{-\lambda n \widehat{L}_t}$.
5. Consider the first-order certificate $\mathcal{B}(\rho) = 2\,\overline{\mathrm{kl}}^{-1}(q(\rho), \varepsilon(\rho))$ with $q(\rho) = \rho^\top \widehat{L}$ and $\varepsilon(\rho) = \frac{\mathrm{KL}(\rho\|\pi) + \ln(2\sqrt{n}/\delta)}{n}$. Admitting the derivatives $\frac{\partial p}{\partial q}$ and $\frac{\partial p}{\partial \varepsilon}$ of $p = \overline{\mathrm{kl}}^{-1}(q, \varepsilon)$ (Exercise 12, question 4), write $\nabla_\theta \mathcal{B}$, and simplify it at $\rho = \pi$.

6. Numerical application: three voters, $\widehat{L} = (0.1, 0.2, 0.3)$, π uniform, $\theta = \mathbf{0}$, $n = 200$, $\delta = 0.05$. Compute ε, $p = \overline{\mathrm{kl}}^{-1}(q, \varepsilon)$ (with `kl_inv_upper`), $\mathcal{B}$, the two partial derivatives and $\nabla_\theta \mathcal{B}$. Perform one gradient step $\theta \leftarrow \theta - 5\nabla_\theta \mathcal{B}$ and compute the new certificate. Where do repeated steps lead?